Bhargav Hegde

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Graduate AI Engineer with 3+ years of software engineering experience, specializing in Machine Learning, Computer Vision, and Autonomous Vehicle Perception Systems

Education

State University of New York at Buffalo, Buffalo, NY Aug. 2024 – Present
Master of Science in Computer Science and Engineering
Research Associate in CAVAS Lab (Connected and Autonomous Vehicle Applications and Systems) - conducting research on real-time 3D object detection, LiDAR-based perception, and V2X cooperative perception for autonomous vehicles

JSS Academy of Technical Education, Bangalore, India Aug. 2016 – Aug. 2020
Bachelor of Engineering in Computer Science and Engineering
Initiated and **secured funding** for an IoT laboratory; conducted instructional sessions for junior students

Professional Experience

Tech Mahindra, Software Engineer Nov. 2021 – Jul. 2024
Developed automation frameworks using Robot Framework, applying machine learning-based pattern recognition for Wi-Fi RTT locationing systems (IEEE 802.11mc), achieving sub-meter accuracy through advanced packet analysis with Wireshark.
Led R&D initiatives in DHCPv6 protocol testing, designed comprehensive test cases addressing stateful vs. stateless configurations, and delivered training to cross-functional teams globally on Wi-Fi RTT implementation.

Tata Consultancy Services, System Associate Engineer Apr. 2021 – Oct. 2021
Developed Python-based RESTful APIs to automate business processes, streamlining data fetching, processing, and storage for efficient client reporting, accelerating deployment time by 15%.

Internships 2018 – 2020
Pentagon Space (2020): Completed 4-month Python Full Stack course; built web applications with Django and SQL.
Teqed Labs (2019): Developed Computer Vision-based attendance system using Deep Learning for face recognition and database management — **Awarded 'Best Project'** for delivering high-impact solution.
Experts Hub (2018): Built machine learning pipeline to classify plant and seedling images with 92% accuracy on 10,000+ image dataset, optimizing species recognition. Led and mentored team of 8 members — **Awarded 'Best Intern'**.

Projects

DAIR-V2X Cooperative LIDAR 3D Object Detection (PyTorch, CUDA, V2X) 2025
Full reproduction of DAIR-V2X late-fusion model for cooperative 3D object detection in V2X scenarios, achieving 40.01% AP on vehicle class with improved performance metrics: 37.25% 3D AP@0.7 (8.9% gain) and 50.51% BEV AP@0.7 (6.7% gain).
Reduced communication cost by 12.3% to 898.10 bytes; implemented LiDAR point cloud processing with visualization tools for 3D and BEV fusion results.

RAG-KnowBot: Personal Knowledge Chatbot (Python, LangChain, Chroma, Llama 3.1) 2025
Built fully local, private Retrieval-Augmented Generation chatbot using Llama 3.1 8B and nomic-embed-text embeddings for interacting with personal documents (PDF, TXT, Markdown).
Features file upload/deletion, customizable prompts, source citations, persistent vector database with Chroma, and hallucination guardrails; runs completely offline for privacy.

Garbage Segregation System (YOLOv3, CNN, Raspberry Pi, Robotic Arm) Jun. 2020
Designed robotic waste segregation system using YOLOv3 and custom CNNs, achieving 90% garbage classification accuracy with optimized edge inference on Raspberry Pi.
Automated waste sorting and placement with integrated robotic arm; **secured funding from Karnataka State Council for Science & Technology (KSCST)**.
For a complete list of projects, please visit: bhargavhegde.github.io

Technical Skills

Technical Skills
Programming Languages: Python, C/C++, Java, SQL, JavaScript, HTML/CSS, Shell Scripting
AI/ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, YOLOv3, CNNs, LangChain, Ollama, ROS2
Deep Learning: Computer Vision, NLP, Reinforcement Learning (SARSA, Q-learning), LLMs (Llama, Mistral), RAG
Specialized Technologies: LiDAR Processing, 3D Object Detection, V2X Communications, CUDA, Embedded Systems (Raspberry Pi)
Tools & Platforms: Docker, Git, Wireshark, Robot Framework, Streamlit, Chroma, Optuna, AWS, JIRA
Databases & Systems: MySQL, SAP, Linux/UNIX, IoT **Networking:** IEEE 802.11mc, DHCPv6, IPv6, Wi-Fi RTT